USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #8 (Lesson 25) - Tapering Lab
Lab Sheet
Fall 2026

Name: ______________________ **Section:** __________ **Date:** ____________

Learning Objective 3.7: Apply amplitude tapering to control sidelobe level, and predict the resulting pattern trade-off.

This sheet is the turn-in document for the Lesson 25 lab. Predict each row before you sweep it, and keep the peak drop and the taper efficiency in separate columns throughout. The lab is a team assignment, so one sheet per team.

1. **The taper table, part 1: the element gains.** Read the eight slider values off Element Gains for each preset and record them as percentages.

Preset	**Rx1**	**Rx2**	**Rx3**	**Rx4**	**Rx5**	**Rx6**	**Rx7**	**Rx8**
Uniform								
Hann								
Blackman								
Chebyshev								

The taper table, part 2: what the sweep reads. Predict each row before you press Start. The peak drop is measured against the frozen uniform trace, and $\eta_t = (\sum a_n)^2/(N \sum a_n^2)$ is computed rather than measured.

Preset	**HPBW pred**	**HPBW meas**	**Drop pred**	**Drop meas**	**First sidelobe**	η_t	η_t **(dB)**
Uniform							
Hann							
Blackman							
Chebyshev							

2. Your custom taper. Record the eight gain values you settled on and the measured result. The specification was a half-power beamwidth no wider than 17° with the first sidelobe below −20 dBc.

Taper	**Rx1**	**Rx2**	**Rx3**	**Rx4**	**Rx5**	**Rx6**	**Rx7**	**Rx8**
Custom								

Measured HPBW = ______ Measured first sidelobe = ______

In one sentence, how did you arrive at these values?

3. Written answers. A short paragraph each.

(a) The tapered sidelobes disappear from the plot, but the beamwidth change is easy to see and easy to measure. Explain why the measurement gives a good number for one and no number at all for the other.

(b) Explain why the peak drop you measured is not the directivity the array lost, and what each number would be used for.

Documentation: ______________________________